

Ian C. Weaver, Ph.D.

[Email](#) | [Website](#) | [GitHub](#)

Education

Ph.D. in Astronomy & Astrophysics

Center for Astrophysics | Harvard & Smithsonian

2020 Jun – 2022 May

Cambridge, MA

- **Advisor:** Dr. Mercedes López-Morales
- **Thesis:** Atmospheric characterization of high-gravity hot Jupiters with ACCESS

A.M. in Astronomy & Astrophysics

Center for Astrophysics | Harvard & Smithsonian

2016 Sep - 2020 May

Cambridge, MA

- **Advisor:** Dr. Mercedes López-Morales

B.S. in Astronomy & Astrophysics, (honors)

University of California, Santa Cruz

2012 Sep - 2016 Jun

Santa Cruz, CA

- **Advisor:** Dr. Enrico Ramirez-Ruiz
- **Honors Senior Thesis:** Modeling Accretion Stream and Disk Evolution in WASP-12/b

Teaching

Program Manager and Tutor Coordinator

Onaketa [\[link\]](#)

2022 May - Present

Oakland, CA

- Coordinate tutor-student matching, provide administrative support, and contribute to growth of non-profit organization.
- Tutor and mentor Black and Brown students in STEM courses through weekly 1-2-1 remote sessions during the academic year.
- Organize and lead student/guardian orientations.
- Process time sheets.
- Generate program and student metric reports.
- Assist in filing county and state reporting data.

PDP Team Leader

ASTRAL Community College Workshop

Summer 2025

Lick Observatory, CA

- Led a three-person team to provide a week-long investigation module as part of the University of California Observatories (UCO) ASTRAL Community College Workshops at Lick Observatory [\[link\]](#).
- Informed by best practices developed by the ISEE Professional Development Program (PDP, [\[link\]](#)), this module taught 12 bay area community college students how to take their own observation of an exoplanet transit, and interpret and share their results using modern tools in astronomy.

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observation of an asteroid occultation, and interpret and share their results using modern tools in astronomy.

Academic Tutor

AJ Tutoring

2022 Aug - 2023 Aug

Palo Alto, CA

- Provided in-person tutoring for high school physics and math courses including AP Calculus AB/BC, and AP Physics 1/2/C Mechanics/C E&M.
- Created custom study materials for students and communicated learning outcomes and progress reports to parents and guardians.

Summer Research Co-Leader

Harvard Banneker Institute Stars Workshop

Summer 2019

Cambridge, MA

- Assisted in the planning and teaching of a week long workshop on stellar evolution for the Banneker Institute as part of the [ISEE Professional Development Program \(PDP\)](#).

Teaching Fellow

Astronomy 16, Stellar and Planetary Astronomy

Spring 2019

Harvard University

- Assisted in course planning, teaching, grading, and lab management for 20+ students.

Workshop Leader

BI, Stellar Evolution

Summer 2018

Harvard University

- Led week-long course in stellar evolution for Banneker Institute scholars.

Teaching Fellow

Astronomy S35, Fundamentals of Contemporary Astronomy

Summer 2018

Harvard University

- Assisted in course planning, teaching, grading, and lab management for 30+ summer bridge high school students.

Teaching Fellow

Astronomy 110, Exoplanets

Fall 2017

Harvard University

- Assisted in course planning, teaching, grading, and lab management for 15+ upper-division undergraduates.

Physics Co-leader

UCSC Academic Excellence Program (ACE) [\[link\]](#)

Fall 2013 - Spring 2016

University of California, Santa Cruz

- Assisted physics section leader in mentoring and tutoring introductory physics students in large 25-30 person sessions for 1 hour and 45 minutes twice a week, and personally five times a week in smaller 4-6 person one hour sessions.

Research

Astronomer, Education Program Lead

SETI - Unistellar [\[link\]](#)

2023 Aug - Present

San Francisco, CA

- Lead for the Unistellar College Astronomy Network (UCAN, [\[link\]](#)), providing free telescopes and educational material to community colleges nationwide.
- Assist in community outreach as part of NASA's upcoming Multi-slit Solar Explorer (MUSE) mission [\[link\]](#).

- Assist in developing education materials and programs for the Hat Creek Radio Observatory (HCRO, [\[link\]](#)) and NASA Community College Network (NCCN, [\[link\]](#)).
- Host star parties in national parks to increase public engagement in astronomy.

Doctoral Student Researcher

Harvard University

2016 Sep - 2022 May

Cambridge, MA

- Provided novel ground-based observations and atmospheric characterizations for the underrepresented class of high-gravity, hot Jupiters.

Undergraduate Student Researcher

UC Santa Cruz

2012 - 2016

Santa Cruz, CA

- Implemented the Adaptive Mesh Refinement code FLASH to apply a full hydrodynamical treatment of accretion stream and disk formation in WASP-12/b. Research performed as part of the CAMP, UC LEADS, and Lamat Scholars programs.
- Developed a novel simulation tracer for modeling mass transfer in WASP-12/b and other binary exoplanetary systems by computing massless particle trajectories in a non-inertial reference frame.

Outreach

Onaketa's Dream Big Book Fair [\[link\]](#)

Organizing Team

Spring 2023, 2024, 2025, 2026

Oakland, CA

- Collected books, set up tables, and cataloged inventory of 1000+ books centered around Black and Brown scientists, engineers, mathematicians, and artists for free giveaway.
- Coordinate communications, marketing, and volunteers.
- Managed ticket sales and guest check-ins for hundreds of participants.

Boys & Girls Clubs of the Peninsula [\[link\]](#)

Planetary ReaCH | SETI

Summer 2023

East Palo Alto, CA

- Designed and co-led hands-on science outreach activities for elementary – high school students in the community as part of the Culturally Inclusive Planetary Engagement Workshop.
- Children designed, built, and tested their own model lunar landers using household materials.

Harvard Observing Project [\[link\]](#)

Lead Observer

Spring 2017 - Summer 2022

Cambridge, MA

- Operated the 0.4m Clay Telescope for weekly star parties open to the public and for student labs.
- Led team of undergraduate students in observing RW Aurigae for a student research project.

ComSciCon [\[link\]](#)

Team Coordinator

Fall 2016 - Spring 2017

Cambridge, MA

- Read and ranked 200+ applications. Handled dining logistics with multi-thousand dollar budget for conference attendees.

GSAS - Open Labs at Harvard [\[link\]](#)

Co-Director

Fall 2016 - Spring 2017

Cambridge, MA

- Co-founded Graduate School of Arts and Sciences (GSAS) Harvard chapter of Open Labs, a science outreach program devoted to sharing graduate student research to 6th-12th grade students through fun, TED style like talks.

Smithsonian Astrophysical Observatory Latino Initiative Program [\[link\]](#)

Python Workshop Instructor

Summer 2017

Cambridge, CA

- Guided Latino Initiative Program scholars through a workshop dedicated to learning important Python based tools in the astronomy community to processes and visualize different types of data.

Observing Experience

Accepted Observing Proposals (PI)

ACCESS: Probing Exoplanet Atmospheres and Enabling TESS Follow-Up with MMT/Binospec

5 nights | 6.5m Magellan Telescope | IMACS 2021A

1 nights | 6.5m Magellan Telescope | IMACS 2020B

4 nights | 6.5m Magellan Telescope | IMACS 2020A

ACCESS-North: Probing Exoplanet Atmospheres and Enabling TESS

3 nights | 6.5m MMT Telescope | Binospec 2019A

3 nights | 6.5m MMT Telescope | Binospec 2018C

2 nights | 6.5m Magellan Telescope | IMACS 2018B

On-site Observing

5 nights | 6.5m Magellan Telescope | IMACS 2017 – 2019

2 nights | 2.4m MDM Telescope | OSMOS 2018B

1 nights | 3m Shane Telescope | Kast 2015B

Open-Source Software Contributions

Core maintainer for the JuliaAstro organization [\[link\]](#), the official organization for astronomy and astrophysics software in the Julia programming language [\[link\]](#).

A collection of most recent contributions to this and other open-source initiatives can be found here [\[link\]](#).

Presentations

Selected Talks

ACCESS: An optical transmission spectrum of the high-gravity, hot Jupiter

Spring 2021

HAT-P-23b

Exoplanet Journal Club (Invited)

University of Chicago

ACCESS: A Flat Visual Spectrum of the Hot Jupiter WASP-43b

Fall 2020

Exoplanetary Science Initiative (ESI) Lecture Series (Invited)

JPL

A New Optical to near-IR Transmission Spectrum of WASP-43b Planetary Astrophysics Seminar Series (Invited)	Winter 2019 Yale
A New Optical to near-IR Transmission Spectrum of WASP-43b Boston Area Exoplanet Science Meeting	Fall 2018 MIT
ACCESS on Magellan: A survey of Optical Transmission Spectra of Exoplanetary Atmospheres Conference on Transiting Exoplanets	Summer 2017 Keele University

Selected Posters

A New Optical to near-IR Transmission Spectrum of WASP-43b Exoplanets II Conference	Summer 2018 Cambridge University, UK
Applying a Hydrodynamical Treatment of Stream Flow and Accretion Disk Formation in WASP 12/b Exoplanetary System AAS 227th Meeting	Winter 2016 Kissimmee, Florida
Particle Trajectory Calculations in WASP-12/b National Society of Black Engineers (NSBE) National Convention	Spring 2014 Philadelphia, Pennsylvania
Mass Transfer in WASP-12 System California Alliance for Minority Participation (CAMP) Symposium	Winter 2014 UC Irvine

Honors and Awards

Derek Bok Center Certificate of Distinction in Teaching	Spring 2019 Harvard University
Derek Bok Center Certificate of Distinction in Teaching	Fall 2017 Harvard University
Chancellor's Award	Spring 2017 UC Santa Cruz
Dean's Award	Spring 2017 UC Santa Cruz
SACNAS Symposium Honorable Mention	Fall 2014
University of California's Leadership Excellence through Advanced Degrees (UC LEADS)	Summer 2014 UC Santa Cruz
National Science Foundation LAMAT Fellowship	Summer 2014 UC Santa Cruz
Ron Ruby Scholarship	Spring 2014 UC Santa Cruz
California Space Grant Consortium Undergraduate Research Opportunity Program (CaSGC)	Spring 2014 UC Santa Cruz
CAMP Symposium Honorable Mention	Winter 2013

Publications

ADS [\[link\]](#).

Allen, N. H., Espinoza, N., Jordán, A., López-Morales, M., Apai, D., Rackham, B. V., Kirk, J., Osip, D. J.,
Weaver, I. C., McGruder, C., Ceballos, K. O., Reggiani, H., Brahm, R., Rodler, F., Lewis, N. K., &

- Fraine, J. (2022). ACCESS: Tentative Detection of H₂O in the Ground-based Optical Transmission Spectrum of the Low-density Hot Saturn HATS-5b. *AJ*, 164(4), 153. <https://doi.org/10.3847/1538-3881/ac8b74>
- Bixel, A., Rackham, B. V., Apai, D., Espinoza, N., López-Morales, M., Osip, D., Jordán, A., McGruder, C., & Weaver, I. (2019). ACCESS: Ground-based Optical Transmission Spectroscopy of the Hot Jupiter WASP-4b. *AJ*, 157(2), 68. <https://doi.org/10.3847/1538-3881/aaf9a3>
- Espinoza, N., Rackham, B. V., Jordán, A., Apai, D., López-Morales, M., Osip, D. J., Grimm, S. L., Hoeijmakers, J., Wilson, P. A., Bixel, A., McGruder, C., Rodler, F., Weaver, I., Lewis, N. K., Fortney, J. J., & Fraine, J. (2019). ACCESS: a featureless optical transmission spectrum for WASP-19b from Magellan/IMACS. *MNRAS*, 482(2), 2065–2087. <https://doi.org/10.1093/mnras/sty2691>
- Kirk, J., López-Morales, M., Wheatley, P. J., **Weaver, I. C.**, Skillen, I., Louden, T., McCormac, J., & Espinoza, N. (2019). LRG-BEASTS: Transmission Spectroscopy and Retrieval Analysis of the Highly Inflated Saturn-mass Planet WASP-39b. *AJ*, 158(4), 144. <https://doi.org/10.3847/1538-3881/ab397d>
- Kirk, J., Rackham, B. V., MacDonald, R. J., López-Morales, M., Espinoza, N., Lendl, M., Wilson, J., Osip, D. J., Wheatley, P. J., Skillen, I., Apai, D., Bixel, A., Gibson, N. P., Jordán, A., Lewis, N. K., Louden, T., McGruder, C. D., Nikolov, N., Rodler, F., & **Weaver, I. C.** (2021). ACCESS and LRG-BEASTS: A Precise New Optical Transmission Spectrum of the Ultrahot Jupiter WASP-103b. *AJ*, 162(1), 34. <https://doi.org/10.3847/1538-3881/abfcd2>
- McGruder, C. D., López-Morales, M., Espinoza, N., Rackham, B. V., Apai, D., Jordán, A., Osip, D. J., Alam, M. K., Bixel, A., Fortney, J. J., Henry, G. W., Kirk, J., Lewis, N. K., Rodler, F., & **Weaver, I. C.** (2020). ACCESS: Confirmation of No Potassium in the Atmosphere of WASP-31b. *AJ*, 160(5), 230. <https://doi.org/10.3847/1538-3881/abb806>
- McGruder, C. D., López-Morales, M., Kirk, J., Espinoza, N., Rackham, B. V., Alam, M. K., Allen, N., Nikolov, N., **Weaver, I. C.**, Ortiz Ceballos, K., Osip, D. J., Apai, D., Jordán, A., & Fortney, J. J. (2022). ACCESS: Confirmation of a Clear Atmosphere for WASP-96b and a Comparison of Light Curve Detrending Techniques. *AJ*, 164(4), 134. <https://doi.org/10.3847/1538-3881/ac7f2e>
- Peluso, D. O., Dalba, P. A., Wright, D., Esposito, T. M., Sgro, L. A., **Weaver, I. C.**, Marchis, F., Dragomir, D., Villanueva, S., Fulton, B., Isaacson, H., Gupta, A. F., Jacobs, T. L., LaCourse, D. M., Gagliano, R., Kristiansen, M. H., Omohundro, M., Schwengeler, H. M., Terentev, I. A., ... Yue, W.-C. (2024). Confirming the Warm and Dense Sub-Saturn TIC 139270665 b with the Automated Planet Finder and Unistellar Citizen Science Network. *AJ*, 167(4), 170. <https://doi.org/10.3847/1538-3881/ad2772>
- Peluso, D. O., Megowan-Romanowicz, C., Pennypacker, C., Marchis, F., Esposito, T., Dalba, P. A., **Weaver, I. C.**, Wright, D., Carter, B., & Sgro, L. (2023). Astronomy Modeling Instruction with Exoplanets & the Unistellar Telescope Network. *AGU Fall Meeting Abstracts, 2023*, ED13A–3.
- Weaver, I. C.**, López-Morales, M., Alam, M. K., Espinoza, N., Rackham, B. V., Goyal, J. M., MacDonald, R. J., Lewis, N. K., Apai, D., Bixel, A., Jordán, A., Kirk, J., McGruder, C., & Osip, D. J. (2021). ACCESS: An Optical Transmission Spectrum of the High-gravity Hot Jupiter HAT-P-23b. *AJ*, 161(6), 278. <https://doi.org/10.3847/1538-3881/abf652>
- Weaver, I. C.**, López-Morales, M., Espinoza, N., Rackham, B. V., Osip, D. J., Apai, D., Jordán, A., Bixel, A., Lewis, N. K., Alam, M. K., Kirk, J., McGruder, C., Rodler, F., & Fienko, J. (2020). ACCESS: A Visual to Near-infrared Spectrum of the Hot Jupiter WASP-43b with Evidence of H₂O, but No Evidence of Na or K. *AJ*, 159(1), 13. <https://doi.org/10.3847/1538-3881/ab55da>